

04.01 Radical Expressions

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
A radical contains a perfect-square factor.	Factor the radicand and take out each perfect-square factor.
Terms have matching simplified radicals.	Combine their coefficients like like terms.
A product or quotient contains radicals.	Apply the product or quotient property, then simplify.

A. Simplify radicals

1.

$$\sqrt{72}$$

2.

$$\sqrt{180}$$

3.

$$\sqrt{98}$$

4.

$$\sqrt{75}$$

5.

$$\sqrt{48}$$

6.

$$\sqrt{200}$$

7.

$$\sqrt{125}$$

8.

$$\sqrt{250}$$

9.

$$\sqrt{45}$$

10.

$$\sqrt{147}$$

11.

$$\sqrt{192}$$

12.

$$\sqrt{108}$$

13.

$$\sqrt{242}$$

14.

$$\sqrt{288}$$

15.

$$\sqrt{320}$$

16.

$$\sqrt{363}$$

17.

$$\sqrt{432}$$

18.

$$\sqrt{500}$$

19.

$$\sqrt{675}$$

20.

$$\sqrt{800}$$

B. Combine like radicals

1.

$$\sqrt{12} + \sqrt{27}$$

2.

$$2\sqrt{18} + \sqrt{8}$$

3.

$$3\sqrt{20} - \sqrt{45}$$

4.

$$\sqrt{50} + 2\sqrt{8}$$

5.

$$4\sqrt{27} - \sqrt{12}$$

6.

$$\sqrt{75} + \sqrt{48}$$

7.

$$2\sqrt{32} - 3\sqrt{2}$$

8.

$$5\sqrt{18} + \sqrt{50}$$

9.

$$\sqrt{98} - \sqrt{8}$$

10.

$$3\sqrt{45} - 2\sqrt{20}$$

11.

$$\sqrt{108} + \sqrt{12}$$

12.

$$2\sqrt{63} + \sqrt{28}$$

13.

$$\sqrt{147} - \sqrt{75}$$

14.

$$4\sqrt{24} + \sqrt{54}$$

15.

$$\sqrt{200} - 2\sqrt{50}$$

16.

$$3\sqrt{80} + \sqrt{45}$$

17.

$$\sqrt{288} - \sqrt{32}$$

18.

$$2\sqrt{125} + \sqrt{20}$$

19.

$$\sqrt{192} + 3\sqrt{12}$$

20.

$$5\sqrt{72} - \sqrt{18}$$

C. Multiply or divide radicals

1.

$$\sqrt{8} \cdot \sqrt{18}$$

2.

$$\sqrt{12} \cdot \sqrt{27}$$

3.

$$\sqrt{20} \cdot \sqrt{5}$$

4.

$$\sqrt{45} \cdot \sqrt{20}$$

5.

$$\sqrt{18} \cdot \sqrt{50}$$

6.

$$\frac{\sqrt{72}}{\sqrt{2}}$$

7.

$$\frac{\sqrt{80}}{\sqrt{5}}$$

8.

$$\frac{\sqrt{98}}{\sqrt{2}}$$

9.

$$\frac{\sqrt{150}}{\sqrt{6}}$$

10.

$$\frac{\sqrt{200}}{\sqrt{8}}$$

11.

$$\sqrt{6} \cdot \sqrt{24}$$

12.

$$\sqrt{15} \cdot \sqrt{60}$$

13.

$$\frac{\sqrt{245}}{\sqrt{5}}$$

14.

$$\frac{\sqrt{288}}{\sqrt{2}}$$

15.

$$\sqrt{14} \cdot \sqrt{56}$$

16.

$$\frac{\sqrt{363}}{\sqrt{3}}$$

17.

$$\sqrt{21} \cdot \sqrt{84}$$

18.

$$\frac{\sqrt{432}}{\sqrt{3}}$$

19.

$$\sqrt{10} \cdot \sqrt{90}$$

20.

$$\frac{\sqrt{500}}{\sqrt{20}}$$